CCC-C-439a
JULY 27, 1959
SUPERSEDING
Int. Fed. Spec. CCC–C–00439 (GSA–FSS)
October 8, 1957 and
Fed. Spec. CCC–B–791a
August 27, 1937

FEDERAL SPECIFICATION

CLOTH, COTTON, BUNTING, MERCERIZED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers cotton bunting primarily intended for use in the manufacture of signal, semaphore, ensign (includes U.S. flag), and other type flags.

1.2 Classification.

1.2.1 *Type and color.*—Bunting covered by this specification shall be of one type and of the color specified in the invitation for bids.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

TT–C–591—Colors; (For) Flag of the United States.

CCC–T–191—Textile Test Methods.

PPP–P–51 — Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks, and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the

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procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Color Association of the United States Inc.
Standard Color Card of America

(Copies of the Standard Color Card of America may be obtained from the Textile Color Card Association of the United States, Inc., 200 Madison Avenue, New York 16, New York.)

3. REQUIREMENTS

3.1 Material.

3.1.2 *Cotton.*—The cotton shall be of suitable grade and staple length to meet the requirements of this specification.

3.1.3 *Yarn.*—The yarn shall be two ply in both warp and filling and shall be made from cotton which has been thoroughly cleaned, well carded, and evenly drawn.

3.1.4 *Finish.* — The cloth shall be fully mercerized and closely singed and shall contain not more than 2 percent of sizing, finishing, or other nonfibrous material when tested as specified in table VI. The pH, when tested as specified in table VI shall be not less than 6 nor more than 9.

3.1.5 *Physical requirements.* — The cloth shall be of plain weave construction and shall have two nonraveling selvages. Physical requirements shall otherwise conform to the requirements of table I.

3.1.6 *Standard sample.*—The cloth shall be equal to the standard sample in all respects not specifically covered by this specification (see 6.3).

3.2 Color.—The color shall be as specified (see 6.1) and shall be produced by such colorants and methods as to meet the fastness requirements of 3.2.1. When the basic colors of the United States flag are required, they shall match cable numbers 70180 (red), 70001 (white), and 70075 (blue) of the Standard Color Card of America, as applicable. The red (70180) may be dyed with naphthol dyes and the blue (70075) shall be dyed with vat dyes.

3.2.1 *Colorfastness.*

3.2.1.1 *Rating per standard sample.*—The dyed bunting shall show colorfastness to crocking, light, water, and salt water and soap equal to or better than the standard sample when tested as specified in table VI.

TABLE I.—*Physical requirements*

Width	Weight per sq. yd. (minimum)	Yarns per inch (minimum)		Breaking strength (minimum)	
		Warp	Filling	Warp	Filling
Inches	*Ounces*			*Pounds*	*Pounds*
18 inch, 36 inch, or as specified (minus tolerance of ¼ inch and plus tolerance of ¾ inch) exclusive of selvage.	4.5	31	30	50	50

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3.2.1.2 *Rating per no-standard sample.*— When no standard sample is available the dyed bunting shall show fair colorfastness to crocking and good colorfastness to light, water, and salt water and soap when tested as specified in table VI.

3.3 Ticket.—Each roll or bolt shall contain an identification ticket which shall conform to the applicable requirements of Federal Specification PPP–P–51.

3.4 Length of roll or bolt.—Unless otherwise specified, any roll or bolt of 35 continuous yards or over will be acceptable.

3.5 Workmanship.—The cloth shall conform to the quality and grade of product established by this specification and the ocurrence of defects shall not exceed the applicable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Civil agencies.—Civil agencies may use sampling, inspection and testing procedures outlined in sub-section 4.2 when desirable. When procedures outlined in 4.2 are not deemed suitable, the procedures specified under civil agency paragraphs shall be followed.

4.1.1 *Sampling.*

4.1.1.1 *Inspection lot.* — Unless otherwise specified, all bunting of the same type submitted at one time shall be considered a lot for purposes of inspection and tests.

4.1.1.2 *Sampling for inspection.*—A random sample of cloth shall be taken from each lot in accordance with table II.

4.1.1.3 *Sampling for tests.*—Random samples for tests shall be taken from each inspection lot in accordance with table III.

4.1.2 *Yard-by-yard inspection.* — Each of the samples selected in accordance with table II shall be visually inspected and defects classified in accordance with table IV. The maximum number of defects acceptable shall be as specified in table II.

4.1.3 *Overall inspection and inspection for length.* — Overall inspection and inspection for length shall be as specified in 4.2.3.2 and 4.2.3.3, respectively. Sample size and the maximum number of defects acceptable shall be as specified in table V.

4.1.4 *Tests.* — Samples selected for tests in accordance with table III shall be tested in accordance with the applicable methods of

TABLE II.—*Sampling for inspection*

Number of yards in lot	Number of rolls or bolts in the inspection sample	Number of yards in inspection sample	Acceptance number, total (defects)	Rejection number, total (defects)	Acceptance number, major (defects)	Rejection number, major (defects)
300 and under	2	35	5	6	1	2
301 to 500	3	50	6	7	2	3
501 to 800	3	75	9	10	3	4
801 to 1,300	3	110	12	13	4	5
1,301 to 3,200	5	150	17	18	5	6
3,201 to 8,000	7	225	24	25	8	9
8,001 to 22,000	10	300	32	33	10	11
22,001 to 110,000	15	450	43	44	14	15
110,001 to 550,000	25	750	68	69	20	21
550,001 and over	50	1,500	124	125	35	36

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TABLE III.—*Sampling for tests*

Number of rolls or bolts in lot	Number of rolls or bolts in sample[1]	Acceptance number (defective)	Rejection number (defective)
8 and under	2	0	1
9 to 15	3	0	1
16 to 40	4	0	1
41 and over	5	0	1

[1] One unit of product (test sample) 3 yards long and the full width of the cloth shall be taken from each sample roll.

Federal Specification CCC–T–191 as listed in table VI of this specification. The physical values specified in section 3 of this specification apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. The unit of product shall be 3 continuous yards full width of finished cloth.

4.1.5 *Compliance criteria.*

4.1.5.1 *Inspection.*—Any piece in the sample for inspection failing to conform to the requirements for which the inspection was made shall be considered defective. If the number of defects in the sample exceeds the acceptance number for that sample specified in table II or V as applicable, the entire lot from which the sample was taken shall be considered to have failed to meet the requirements of this specification.

4.1.5.2 *Tests.*—Any piece in the sample for tests failing to conform to the requirements for which tests were made, shall be considered defective. If the number of defective pieces in the sample exceeds the acceptance number for that sample, the entire lot from which the sample was taken shall be considered to have failed to meet the requirements of this specification.

4.2 **Military agencies.**

4.2.1 *General.*—The quality assurance provisions contained in this section are applicable to sampling, inspection, and testing by and for the Government. Sampling, inspection, and testing of components or materials described shall be in accordance with this specification and with subsidiary specifications referred to herein to the extent applicable, except this specification shall govern in the event of conflict.

4.2.2 *Sampling.*—Sampling shall be in accordance with the provisions of Military Standard MIL–STD–105 and Appendix except where otherwise indicated hereinafter.

4.2.3 *Inspection.*

4.2.3.1 *Yard-by-yard inspection.*—The unit of product for this inspection shall be one linear yard. The inspection level shall be level II of Military Standard MIL–STD–105, and the acceptable quality levels shall be 1.5 majors and 6.5 total defects per 100 units (yards). Defects found during inspection shall be classified in accordance with table IV. The Government reserves the right to require examination for any defect not included herein and to classify such defects in accordance with the definition contained in Military Standard MIL–STD–105. The number of sample pieces (rolls or bolts) over which the sample yardage shall be distributed shall be in accordance with table V. The defects listed in table IV shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs.

4.2.3.2 *Overall examination.*—Each defect listed shall be counted no more than once in each piece examined. The unit of product for this examination shall be one bolt or roll. The sample size (number of bolts or rolls) for this examination and the maximum number of defects acceptable shall be shown in table V.

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TABLE IV.—*Classification of defects*

Glossary No. of Fed. Std. No. 4	Defects	Classification	
		Major	Minor
78	Smash	X	
44, 74, 75, 76	Spots, stains, streaks:		
	Clearly visible and extending over more than 1 inch in combined dimensions	X	
	Clearly visible but extending over 1 inch or less in combined dimensions		X
90, 91, 92	Holes, cuts, tears—Any size	X	
5, 7, 29	Slubs, knots, bunch, etc.—More than 3 times the normal diameter of the yarn	X	
40	Wrong draw—Nine inches or more in length		X
59	Hard crease—Embedded, clearly noticeable		X
11, 33	Broken or missing yarn:		
	Two or more contiguous, ½ inch or more missing, or single, more than 4 inches missing	X	
	Single, 4 inches or less missing		X
13, 34	Float:		
	Any multiple, or single, floating, over more than 4 inches	X	
	Single floating 4 inches or less		X
4, 10, 13	Mispick—Resulting in harness skips or in double pick with more than two yarns in shed	X	
26, 31	Coarse or fine end (multiple)—More than ¼ yard long		X
14	Shade bar, mixed filling—Clearly noticeable		X
12, 35	Open place, crack or hitch-back—Clearly noticeable	X	
79, 89	Abrasion mark, bruise, tender or weak spot	X	
9	Fine or light filling bar—Clearly noticeable		X
30	Hanging or loose threads		X
36	Mixed end		X
57	Finishing bar		X
70	Dye streak	X	
41	Reed mark—Clearly noticeable		X
38	Slack end—Two or more contiguous, jerked-in between picks, or forming clearly noticeable loop on surface of cloth	X	

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TABLE IV.—*Classification of defects* (Cont'd)

Glossary No. of Fed. Std. No. 4	Defects	Classification	
		Major	Minor
37	Tight end:		
	Multiple or any tight section causing waviness in adjoining area	X	
	Single, causing noticeable waviness in adjoining area or forming sharp ridge in cloth		X
1, 2	Coarse filling bar—coarse filling, heavy place—Clearly noticeable	X	
1, 26, 31, 8, 25, 47	Coarse, fine or uneven yarn, clearly visible at 6 feet distance		X
89, 3	Fine or light filling bar, light place, clearly noticeable	X	
77, 5, 27	Gout, slubs, slugs, knots—more than three times the thickness of normal yarn		X
3, 5	Sloughed-off, jerked in filling:		
	3 or more additional yarns in shed	X	
	2 additional yarns in shed		X

TABLE V.—*Sampling for inspection (Military)*

Lot size in yards	Sample size in rolls or bolts	Accetance[2] number (defects)
Up to 1,300[1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	1
110,001 to 550,000	25	1
550,001 and over	50	2

[1] If lot contains fewer than 3 pieces, each piece in the lot shall be examined.

[2] Applies to paragraph 4.2.3.2 and 4.2.3.3 only.

Defects

(*a*) Width less than specified.

(*b*) Baggy, ridgy or wavy cloth, clearly noticeable.

(*c*) Selvage curled, doubled, folded, rolled, slack, tight, cut, broken or torn intermittently, continuously over more than 20 yards of the piece.

(*d*) Off shade—not within the established tolerance.

(*e*) Non-uniform in shade—shaded end-to-end, side-to-side, side-to-center, mottled, or poor dye penetration, two sidedness.

(*f*) Overall uncleanliness.

(*g*) Objectionable odor.

(*h*) Neppiness, clearly noticeable throughout.

(*i*) Not uniformly woven throughout.

4.2.3.3 *Inspection for length.*—Individual rolls shall be considered defective in regard to length requirements if the actual gross length of any one piece is less than 35 yards, or more than 2 yards shorter than the total gross length marked on the piece ticket. The lot shall be unacceptable if the total of the actual gross length of pieces in the sample is less than the total of the gross lengths marked on the piece tickets. The sample size

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(number of rolls or bolts) and maximum number of defects acceptable shall be as shown in table V.

4.2.4 *Tests.*—The methods of testing specified in Federal Specification CCC–T–191, wherever applicable, and as listed in table VI shall be followed. The physical values specified in section 3 of this specification apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. The unit of product for test purposes shall be 3 continuous yards full width of finished cloth. The sample size shall be in accordance with Level L1, table VII of Appendix to Military Standard MIL–STD–105. The acceptable quality level shall be 4.0 per hundred units.

TABLE VI.—*Test methods*

Characteristics	Requirement reference	Method No.
Nonfibrous material	3.1.4	2611
Weight, ounces per square yard	Table I	5041
Identification of cotton	3.1.3	1200
Yarns per inch	Table I	5050
Breaking strength	Table I	5100
Ply	3.1.3	Visual
Weave	3.1.5	Visual
pH	3.1.4	2811
Colorfastness to:		
Crocking	3.2.1	5651
Light[1]	3.2.1	5660[2]
Water	3.2.1	5630
Salt water and soap	3.2.1	5632
Width of cloth	Table I	5020
Length of cloth	4.2.3.3	5010

[1] In the event of dispute resulting from the test with method 5660, or as a result of suspected anomalous behavior of certain dye types or formulation, the contracting officer may authorize the exposure to natural light in accordance with method 5662.

[2] Except that Munsell value shall be not lower than 8.5.

5. PREPARATION FOR DELIVERY

(Civil agencies.—Federal Standard No. 102 shall be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment).

5.1 Packaging.

5.1.1 *Levels A, B, and C.*—The cloth shall be packaged in accordance with the applicable requirements for class 3 fabrics of Federal Specification PPP–P–51.

5.2 Packing.

5.2.1 *Levels A, B, and C.*—The cloth shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51.

5.3 Marking.—Shipments shall be marked in accordance with Federal Specification PPP–P–51.

6. NOTES

6.1 Ordering data. — Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Width (3.1.5).
(*b*) Color (3.2).
(*c*) Length (3.4).
(*d*) Sampling, inspection and test procedures, civil agencies (4.1).
(*e*) Inspection lot, civil agencies (4.1.1.1).
(*f*) Both the level of packaging and the level of packing required (see 5.1 and 5.2).
(*g*) Whether standard sample for comparable colorfastness tests is available (see 3.2.1).

6.2 In the preparation of contracts or orders, it should be noted that various options, choices, and alternatives, as indicated in Federal Specification PPP–P–51, may be exercised in the preparation for delivery of this commodity.

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6.3 Standard sample.—For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Transportation description.—Transportation description applicable to this item is:

Dry goods not otherwise indexed by name:

Carload minimum weight — 20,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor volume minimum weight — any quantity.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—Q
Navy—S MC
Air Force.

Copies of this specification may be purchased for 5 cents each.

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